Problem 1 – Question 1

Axis-angle form of a single-qubit rotation

Let

$$A = \mathbf{n} \cdot \boldsymbol{\sigma} = n_X X + n_Y Y + n_Z Z, \qquad \|\mathbf{n}\| = 1.$$

The Pauli matrices satisfy

$$\sigma_j \sigma_k = \delta_{jk} \mathbb{I} + i \sum_{\ell} \epsilon_{jk\ell} \sigma_\ell, \qquad \{\sigma_j, \sigma_k\} = 2\delta_{jk} \mathbb{I}.$$

Consequently,

$$\begin{aligned} A^2 &= \sum_{j,k} n_j n_k \sigma_j \sigma_k \\ &= \left(\sum_j n_j^2 \right) \mathbb{I} + i \sum_{j,k,\ell} n_j n_k \epsilon_{jk\ell} \sigma_\ell \\ &= \mathbb{I}. \end{aligned}$$

The second term vanishes because $n_j n_k$ is symmetric in j, k, whereas $\epsilon_{jk\ell}$ is antisymmetric. It follows that

$$A^{2m} = \mathbb{I}, \qquad A^{2m+1} = A \quad (m \geq 0).$$

Now expand the exponential and separate its even and odd powers:

$$\begin{aligned} R_{\mathbf{n}}(\theta) &= \exp\left(-i \frac{\theta}{2} A \right) \\ &= \sum_{m=0}^{\infty} \frac{(-i\theta/2)^{2m}}{(2m)!} A^{2m} + \sum_{m=0}^{\infty} \frac{(-i\theta/2)^{2m+1}}{(2m+1)!} A^{2m+1} \\ &= \left[\sum_{m=0}^{\infty} \frac{(-1)^m (\theta/2)^{2m}}{(2m)!} \right] \mathbb{I} - i \left[\sum_{m=0}^{\infty} \frac{(-1)^m (\theta/2)^{2m+1}}{(2m+1)!} \right] A \\ &= \cos\left(\frac{\theta}{2} \right) \mathbb{I} - i \sin\left(\frac{\theta}{2} \right) A. \end{aligned}$$

Substituting the definition of A gives the required expression:

$$\boxed{R_{\mathbf{n}}(\theta) = \cos\left(\frac{\theta}{2} \right) \mathbb{I} - i \sin\left(\frac{\theta}{2} \right) (n_X X + n_Y Y + n_Z Z).}$$